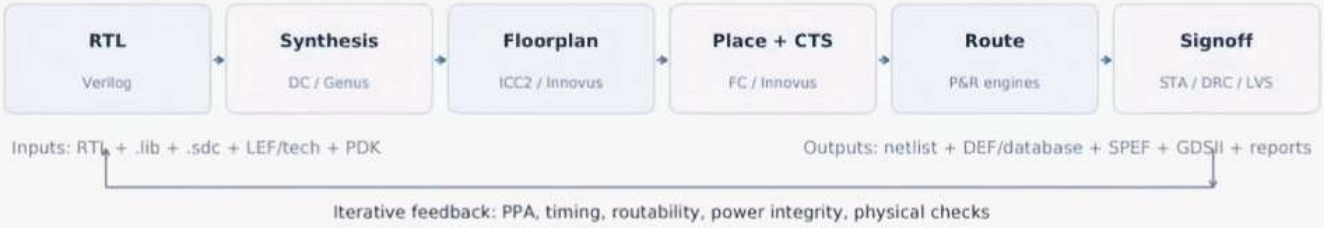
VLSI PHYSICAL DESIGN TOOLS

COMPLETE PRACTICAL & INDUSTRY GUIDE | From Beginner to Industry-Ready Physical Design Engineer



END-TO-END IMPLEMENTATION

Not all steps use separate executables

WHAT PHYSICAL DESIGN ACTUALLY DOES

Register-transfer level (RTL) expresses behavior and register boundaries; physical design (PD) turns a technology-mapped netlist into an electrically and geometrically implementable layout. The layout must satisfy power, performance and area (PPA), manufacturability and reliability across operating conditions.

Frontend covers architecture, RTL, functional verification and often logic synthesis. Backend covers floorplan, physical optimization, clocks, routing and signoff. Boundaries differ by team; logical equivalence guards RTL-to-netlist transformations.

STAGE → INPUTS → ENGINEERING DELIVERABLE

Synthesize	RTL + .lib + SDC → mapped .v, constraints and QoR reports.
Initialize	Tech/LEF, netlist, macros, timing views → design database.
Floorplan	Die/core bounds, pin positions, macros and keepouts → DEF or database.
Power plan	Supply connectivity + rules → rings, straps, rails; PG checks.
Place	Cells + rows + timing targets → legalized instances and congestion map.
CTS	Clock constraints + tree cells → buffered clocks, propagated latency.
Route	Net connectivity + routing rules → connected wires/vias, route DRC.
Extract	Final geometry + process corner → parasitic R/C in SPEF or DSPF.
Validate	STA, power/EMIR, DRC/LVS → signoff reports and final GDSII.

Files to recognize: .lib timing/power model; .lef abstract physical geometry; .sdc constraints; .def placed/routed design interchange; .spef interconnect parasitics; .gds mask-layout data.

Handoff discipline: record the precise RTL/netlist hash, block hierarchy, clock definitions, macro abstracts and release status. A passed report from another database revision is not evidence for the current GDS.

EDA TOOL ROLES — ALTERNATIVES VS PARTNERS

Synopsys	Design Compiler: synthesis; Fusion Compiler: integrated synthesis-to-route; IC Compiler II: physical implementation.
Synopsys signoff	PrimeTime: STA/SI; StarRC: extraction; IC Validator: DRC/LVS; Formality: formal equivalence.
Cadence	Genus: synthesis; Innovus: place/CTS/route; Tempus: STA; Quantus: extraction; Voltus: power/EMIR; Conformal: equivalence.
Siemens EDA	Calibre: physical verification (including DRC and LVS; options vary).
Open ecosystem	Yosys: synthesis; OpenROAD: physical flow; OpenSTA: timing engine; KLayout/Magic: layout inspection/checks.

Compare correctly: DC vs Genus are synthesis alternatives; ICC2/FC vs Innovus overlap for implementation. PrimeTime and StarRC are complementary, not competing functions. Calibre/IC Validator target verification; formal-equivalence tools compare functional models.

THE ACTUAL PROJECT DEPENDS ON THE PDK

- Technology: process-design kit (PDK), layer rules, standard-cell and macro characterization, permitted RC corners, signoff decks.
- Timing modes: clocks, uncertainty, I/O budgets, exceptions, operating voltages/temperatures and on-chip variation.
- Quality gates: setup/hold, fanout/slew/capacitance, congestion, DRC/LVS, IR drop and electromigration (EM).

Operational rule: An EDA result is credible only with the exact library/constraint/corner versions, tool build, source revision and run configuration recorded.

STAGE DEPENDENCIES THAT CAUSE LOOPS

Placement ↔ synthesis: physically poor logic may need restructuring. CTS ↔ hold: balancing can create short-path failures. Route ↔ timing: wire parasitics change criticality. PDN ↔ congestion: strong straps cost routing capacity.

Signed netlist/constraints, layout database and GDS, timing/extraction corners, physical decks and reports, open-issue/waiver ledger, tool-version manifest, screenshots and reproducible run scripts.

LOGIC SYNTHESIS: DC, GENUS & FUSION

Elaboration, mapping, timing constraints, report reading and implementation handoff




```
Illustrative setup check:  required 0.90 ns  -  arrival 1.08 ns  =  slack -0.18 ns
```

Interpretation: delay constraint is violated; inspect cell delay, fanout, wire estimate and SDC.

LOGIC TRANSFORMATION

Numbers and mapped gates above are illustrative

FROM BEHAVIORAL RTL TO CELLS

Elaboration resolves parameters, generate blocks, hierarchy and inferred state. Optimization simplifies logic, shares/restructures expressions and may resize or restructure datapaths. Technology mapping chooses cells from a Liberty library using timing, area and electrical limits.

A netlist is still not a physical layout: gate coordinates, power distribution, clock delivery and routed interconnect are unknown. Post-route delay can differ materially from wire-load or placement estimates.

CONSTRAINTS BEFORE OPTIMIZATION

SDC / CONCEPTUAL EXAMPLE

```
create_clock -name clk -period 2.0 [get_ports clk]
set_input_delay 0.25 -clock clk [get_ports din]
set_output_delay 0.30 -clock clk [get_ports dout]
```

Setup checks maximum data delay before capture; hold checks minimum delay after capture. Slack = required time − arrival time for typical reported max/setup paths; hold reports apply min-path arrival/required conventions. Verify both clocks and exceptions.

TYPICAL DESIGN COMPILER RUN

DC_SHELL; ILLUSTRATIVE LEGACY DC STYLE

```
read_verilog {top.v block.v}
current_design top
link
source constraints.sdc
compile_ultra
report_area
report_timing -max_paths 5
```

Real runs also configure target/link libraries and operating corners; export netlist/SDC/reports via release-specific commands. Do not use a bare fragment as a tapeout-ready flow.

WHAT TO AUDIT AFTER SYNTHESIS

- Unmapped cells/latches: unintended inference, missing libraries, black boxes or unresolved references.
- Constraint health: unconstrained paths and inappropriate false/multicycle exceptions must be reviewed.
- QoR: critical-path depth, sequential/combinational counts, area, cell mix, high-fanout nets, power estimates.

Interview insight: a large synthesis area increase can be driven by over-tight clocks, replicated fanout drivers, inappropriate path exceptions or missing physical context.

HOW TO READ A TIMING REPORT

ILLUSTRATIVE REPORT, NOT TOOL OUTPUT

```
Startpoint: u_reg0/Q  (launch FF)
Endpoint:   u_reg1/D  (capture FF)
Path Group: clk    Type: max
data arrival time       1.08 ns
data required time      0.90 ns
slack (VIOLATED)       -0.18 ns
```

A negative setup slack of 0.18 ns means the data arrives 180 ps late under the stated timing assumptions. Inspect constraints, clock latency/uncertainty, logic depth, net capacitance, fanout and slow-cell choices before changing implementation.

COMMERCIAL TOOL COMPARISON

Design Compiler	Dedicated logical/physical-aware synthesis; common dc_shell command style.
Cadence Genus	Logic synthesis with Cadence flow integration; native command interface differs; consult exact build.
Fusion Compiler	Unified Synopsys synthesis and physical-optimization environment; methodology/command availability vary by release.

Tool-independent handoff: mapped netlist, constraints, hierarchy, library versions, scan/DFT requirements and reproducible timing/area reports. The receiving PD owner must confirm all black-box and macro interfaces.

TRADE-OFF AND FIRST-LEVEL DEBUGGING

- Area vs timing: stronger cells and duplication may improve timing but increase area/leakage; excessive downsizing increases delay.
- Power: toggling capacitance/frequency drives dynamic power; leakage depends on cell threshold, state and PVT.
- Warning: a clean synthesis WNS is not proof of post-route timing. Wire parasitics, CTS and variation are still evolving.

Exercise: Synthesize a counter and multiplier separately; compare mapped cell count, delay, worst path and utilization after placement using the same library and constraints.

FLOORPLAN, POWER PLAN & PLACEMENT

Core geometry, macro channels, PDN design, legalization and congestion diagnosis





CHIP PLANNING + HEATMAPS

Schematic only; heat colors indicate relative demand

INITIALIZE AND DEFINE GEOMETRY

Import the netlist, timing corners, RC tech/LEF, cell and macro abstracts, MMMC views and physical rules. Die bounds the chip; core contains legal standard-cell rows. Approximate utilization = placed standard-cell area ÷ available placement area; macro/obstacle treatment must be stated.

A 65% target is not a universal best practice. Timing, macro mix, pin access, power straps, blockages and router capacity determine feasible density.

MACRO PLACEMENT METHODOLOGY

- Cluster communicating macros; orient pins toward partners; reserve bus corridors and avoid narrow channels.
- Apply halos/keepouts and placement blockages, including clock/power access around edges.
- Check pin-access hot spots before standard-cell placement; do not pin macros solely to minimize estimated wirelength.

PDN AND CELL PLACEMENT

Power delivery network (PDN): rings, vertical/horizontal straps, macro hooks and row rails distribute VDD/VSS. Validate continuity, via access and preliminary IR budget before optimizing timing.

Global placement minimizes estimated wirelength/timing/congestion; legalization moves cells onto valid rows without overlap; detailed placement improves local legality and access. Timing-driven placement can favor shorter critical paths.

PRACTICAL FLOORPLAN GATE

- Die/core, aspect ratio, utilization and macro orientations approved.
- Power domains, PG pins and rails consistent with the UPF/intent.
- Placeable area, macro keepouts, clock sources and I/O pin positions validated.

QUICK UTILIZATION EXAMPLE

If a design contains 42,000 µm² of placeable cell area and the available row area is 70,000 µm², utilization is 60%. Exclude blockages/macros correctly and separately track local hot spots.

DIAGNOSIS: CONGESTION VERSUS DENSITY

Congestion is routing demand divided by available track/capacity resources in a region, not just cell occupancy. A 50% utilized block can still have a macro-pin or narrow-channel routing choke point.

Symptom	High overflow near one macro edge; long detours and growing signal delay.
Verify	Compare placement heatmap, global-route estimates, macro pin density, blockages and wire direction.
Fix	Increase channel width, spread high-fanout clusters, shift or rotate macro, improve pins/straps; re-run place + route estimate.

TOOL OPERATIONS (VERSION AWARE)

CONFIRM CURRENT TOOL COMMAND HELP

```
# ICC2/Fusion: common stage invocations
place_opt
report_timing
check_legality
# Innovus: representative classic UI
place_design
report_timing
```

Some flows use integrated optimization variants (e.g., place_opt_design in Innovus). Never substitute command names mechanically between vendors. Import and power-grid setup is especially methodology-specific.

COMMON FLOORPLAN MISTAKES

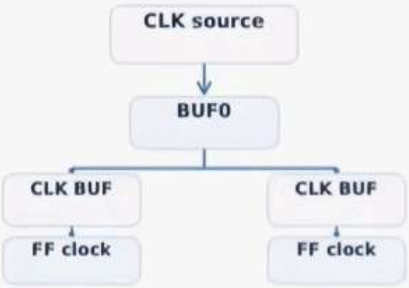
- Overlooking macro PG/pin-side connectivity; too narrow channels; excessive utilization in one local region.
- Treating green congestion as timing closure or assuming placement DRC implies final-route DRC.
- Blindly resizing setup paths: higher drive can increase local congestion and leakage.

Deliverable: Export placed database/DEF, power-connectivity checks, utilization by region, congestion map, top timing paths and a floorplan decisions log.

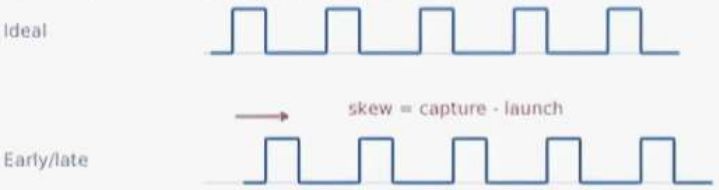
Physical validation: preview placement and global-route feasibility before freezing macro channels or adding dense PG straps.

CLOCK TREE SYNTHESIS & OPTIMIZATION

Latency, skew, uncertainty, clock gating and post-CTS verification



CLOCK NETWORK



Waveform sketch is conceptual, not to physical scale

WHY CLOCK DISTRIBUTION IS SPECIAL

Before clock tree synthesis (CTS), a clock may be treated as ideal or modeled with estimated latency. After CTS, a network of buffers/inverters drives sequential clock pins; propagated clocks model actual path delay. Balance improves robustness but zero skew is neither always possible nor optimal.

Insertion delay: clock-source-to-sink travel time. Skew: capture clock arrival minus launch arrival for a specific relationship. Uncertainty reserves margin for jitter and modeling variation. Both setup and hold depend on launch/capture timing.

PHYSICAL TARGETS AND TRADE-OFFS

- Control sink transition, capacitance, fanout, tree power and clock cell placement.
- Use proper clock-gating recognition and check gated-clock enable paths and generated clocks.
- Useful skew: selectively shifts capture time to aid setup, but may worsen hold or another mode/corner.

CTS EQUATIONS (SIMPLIFIED)

RELATED EDGE / EQUAL PERIOD EXAMPLE

```
Tsetup_margin ~ Tclk + skew
  - uncertainty - tCQ - tDATA - tSETUP
Thold_margin  ~ tCQ_min + tDATA_min
  - skew - tHOLD - hold_uncertainty
```

These intuitive equations omit explicit source-latency, CRPR and multi-cycle terms. Real signoff uses per-edge launch and capture clocks, derates and clock reconvergence pessimism removal (CRPR).

TYPICAL IMPLEMENTATION STEPS

- Confirm clock definitions and exceptions, clock-gating cells and sink groups.
- Choose clock buffers, routing rules/layers and latency/skew objectives.
- Build CTS; inspect skew, insertion delay, transitions, clock-tree depth and area.

NUMERICAL SKEW EXAMPLE

If capture clock arrives at 0.23 ns and launch at 0.15 ns, skew = +0.08 ns. This can help setup on that path, but reduce its hold margin; it is not universally beneficial across endpoints.

WHAT CHANGES AFTER CTS

Clock propagation can transform pre-CTS timing: a capture-late clock helps setup but hurts hold on same-edge paths; a capture-early clock tends toward the opposite. Check both max and min corners with updated clocks and clock reconvergence treatment.

REPORTS: READ BEYOND ONE NUMBER

Insertion latency	Compare source and network components per sink/group, not only mean delay.
Skew distribution	Check local, endpoint-pair and worst skew, not just global skew headline.
Electrical	Inspect clock slew/cap/fanout, buffer usage and gate-enable checks.
Clock power	Track inserted buffers, switching power and placement perturbation.

COMMON COMMANDS — VERIFY RELEASE

ILLUSTRATIVE VENDOR OPERATIONS

```
# Synopsys ICC2/Fusion
clock_opt
report_clock_timing  ;# syntax varies
# Cadence Innovus (classic flow)
ccopt_design
report_ccopt_clock_trees
```

THREE REAL DEBUGGING PATTERNS

Large skew	Check asymmetric routing, cell placement, sink loads, macro detours and intended useful-skew settings.
Many hold fails	Investigate short data paths, changed capture latency, fast corner and overly aggressive setup optimization.
Gating violations	Validate generated clocks, enable timing, integrated clock gating (ICG) arc model and exception scope.

Signoff thinking: Fixing one tree branch is not closure until setup, hold, SI, DRC, clock-gating checks and all active timing views are re-evaluated.

Also audit clock reconvergence, propagated-clock activation, duplicate clock definitions and uncertainty being counted consistently.

ROUTING, SIGNAL INTEGRITY & DESIGN RULES

Global/detail routing, RC penalties, cross-coupling, antenna and DRC repairs

MULTI-LAYER ROUTING



COUPLED NETS / NOISE



Coupling C changes delay and can induce noise.

ROUTING RESOURCES + COUPLING

Actual layer direction and spacing are technology-specific

ROUTING STAGES AND RESOURCES

Global routing assigns tentative routing corridors and exposes regional overflow. Detailed routing chooses tracks, vias and shapes that must obey manufacturing and electrical rules. Routing layers have direction preferences, minimum width/spacing and via restrictions defined by the PDK.

Congestion overflow indicates demand exceeding estimated capacity; a locally routed net may still detour despite a good global summary. Pin access, stacked vias, density rules and macro boundaries complicate legalization.

SIGNAL INTEGRITY / CROSSTALK

Nearby nets share coupling capacitance. An aggressor transition can alter victim delay or induce a glitch; switch alignment and logical state matter. SI-aware timing uses coupling and aggressor timing windows instead of assuming all nets are quiet.

Antenna checks limit damaging charge accumulation during fabrication. Fixes may involve diode cells, layer hopping or approved router repair, subject to rule deck and methodology.

ROUTE DEFECTS: ENGINEER'S TAXONOMY

- Short: unintended conductor connection. Open: disconnected intended terminal or net segment.
- Spacing/width: geometry violates technology minimums; cut/via enclosure may also fail.
- Post-route setup degradation: RC and detours increase arrival time; new coupling can worsen max or min timing.

WORKFLOW: LOCATE BEFORE REPAIRING

Start from exact error coordinates and net; cross-probe DRC marker → layer → shape → pin/access point. Check if repeated markers share a macro edge or one congested region. Repair most systemic issue first; rerun affected route and recheck timing/DRC.

A SIMPLE OVERFLOW THOUGHT EXPERIMENT

A global-route bin with demand 120 track units and capacity 100 has 20 units overflow. Reducing cell utilization alone may not fix the blocked tracks if the cause is a macro edge or dense PDN.

P&R TOOL-STAGE OPERATIONS

COMMAND NAMING IS RELEASE-SENSITIVE

```
# ICC2 / Fusion typical
route_auto
route_opt
# Innovus representative classic command
routeDesign
  # or route_design in newer UI
```

Route optimization can include incremental rerouting, buffering, sizing, layer assignment and via improvement; available controls are tool, license, PDK and methodology dependent.

DEBUGGING EXAMPLES

Overflow hot spot	Verify macro channel, excessive stripes/blockages or localized demand. Adjust geometry/placement before repeatedly rerouting.
Too many vias	Investigate detours, forced layers, forbidden via stacks or fragmented tracks; check via resistance and yield rules.
Slow critical net	Separate cell delay from extracted net delay. Try improved locality, buffering, legal routing layers or cleaner placement.
Antenna markers	Check router antenna-fix pass and signoff rule deck; insert authorized fixes and verify new capacitance.

CRITICAL NET TREATMENT

- Shielding can reduce coupling but consumes tracks and increases capacitance; reserve for justified clocks/sensitive nets.
- Nondefault rules (NDR), wire widening and preferred layers require PDN/DRC and extraction consistency.
- Always check incremental changes against hold, EM, area and route DRC, not only setup WNS.

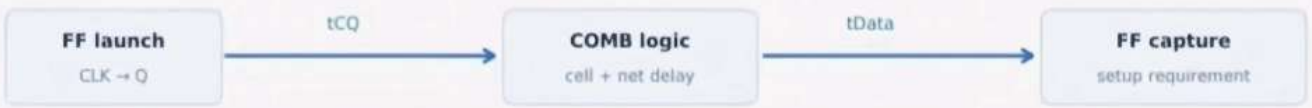
Deliverable: Routed design database, connectivity/DRC reports, congestion map, antenna status, routed timing and extraction-ready metal geometry.

During signoff, electrical violations (max transition/capacitance) are separate from geometric DRC markers; repair priorities depend on severity and tapeout gates.

STATIC TIMING ANALYSIS: PRIMETIME & TEMPUS

Setup, hold, MMMC, derating, exceptions, critical paths and timing ECOs

SETUP PATH (illustrative, same-edge FF to FF)



Required = capture edge + capture latency - setup - uncertainty

Slack = required - arrival

REGISTER-TO-REGISTER SETUP PATH

Clock edge/latency assumptions must be explicitly stated

STA MODELS TIMING WITHOUT SIMULATION

Static timing analysis (STA) propagates cell and interconnect delays through a graph of pins/arcs under stated clocks and operating conditions. It analyzes worst permissible paths rather than testing data vectors; functional correctness still needs verification.

Maximum-delay setup and minimum-delay hold checks are independent. WNS = worst negative slack; TNS = total negative slack over the chosen path/endpoint accounting rule; also track violating endpoints and path groups.

NUMERIC SETUP EXAMPLE

ILLUSTRATIVE SINGLE-CYCLE ANALYSIS

```
Clock period                        1.00 ns
Capture clock latency               0.10
- setup time                       -0.08
- uncertainty                      -0.04
Required time                       0.98
Launch clock 0.08 + tCQ 0.12 +
  logic/wire 0.87 = arrival         1.07
Slack = 0.98 - 1.07                -0.09 ns
```

CRPR, input/output path budgets, actual edge relationships and derates can change the arithmetic. A hold failure requires data arrival too early relative to its minimum allowable time; arbitrary setup buffering may make it worse.

MMMC AND UNCERTAINTY

Multi-mode multi-corner (MMMC) views combine functional/test modes, process-voltage-temperature (PVT) and RC corners. On-chip variation (OCV) models spatial/temporal uncertainty; advanced OCV (AOCV) uses context/depth; parametric OCV (POCV) uses statistical variation models.

Graph-based analysis (GBA) may be pessimistic due to merging possibilities; path-based analysis (PBA) re-evaluates specific paths and can refine criticality.

TIMING WAVEFORM MENTAL MODEL

For setup, sample data before the capture clock edge by the setup aperture; for hold, keep data stable after that edge for the hold aperture. The early or late arrival of propagated clock changes these windows.

READ BOTH MAX AND MIN PATH

ILLUSTRATIVE PAIR, DIFFERENT ANALYSES

```
SETUP: required 0.98 - arrival 1.07
       slack = -0.09 ns (VIOLATED)
HOLD:  arrival 0.15 - required 0.21
       slack = -0.06 ns (VIOLATED)
```

For the above hold illustration, launch-to-capture minimum-delay relation makes 0.21 ns the minimum acceptable arrival. It is not the inverse of the max/setup calculation.

USEFUL TIMING COMMANDS

CHECK OFFICIAL COMMAND HELP

```
# PrimeTime (common pt_shell)
check_timing
report_timing -delay_type max -max_paths 10
report_timing -delay_type min -max_paths 10
report_constraint -all_violators
# Tempus: report_timing is typical;
  # flags/view selection differ by version
```

CONSTRAINT EXCEPTIONS ARE HAZARDOUS

- False path: entirely excluded from selected timing checks; prove impossibility or intent, not merely difficulty.
- Multicycle: modifies launch/capture edge relationship; setup and hold constraints must be paired correctly.
- I/O delays: model external timing budgets. Missing input/output delay can hide real system paths.

A REPEATABLE ECO DECISION SEQUENCE

1. Validate	Check clock, path type, corner, uncertainty, derates, exceptions and path sensitization.
2. Decompose	Cell vs wire delay; fanout, slew, capacitance; launch/capture network.
3. Repair	Move/resize critical gates, buffer high-RC nets, reduce logic depth or alter legal skew.
4. Recheck	Re-extract changed routes; verify setup + hold across all views; record WNS/TNS and side effects.

Check unconstrained endpoints, disabled timing arcs and derived clocks before believing WNS=0. Clean metrics with incomplete timing coverage are misleading.

PARASITIC EXTRACTION & PHYSICAL VERIFICATION

StarRC, Quantus, Calibre, IC Validator, RC-aware STA, DRC and LVS

SPEF-LIKE INTERCONNECT RC MODEL



Rnet and Cground + Ccoupling drive delay/noise.

PHYSICAL VERIFICATION: DIFFERENT CHECKS

DRC	LVS	ERC
geometry	connectivity	electrical

DRC pass does not prove netlist equivalence.
Extraction is separate from LVS and DRC.

INTERCONNECT MODEL + PHYSICAL CHECKS

RC sketch is lumped; signoff uses extracted distributed networks

PARASITIC EXTRACTION CHANGES THE ANSWER

Wires have resistance (R), ground capacitance and coupling capacitance (C); the resulting distributed network affects transition, cell input slew, timing and SI. Pre-route estimates differ from geometry-based post-route extraction and may miss local detours/vias.

Standard Parasitic Exchange Format (SPEF) communicates per-net RC to STA. Signoff extraction requires consistent layer mappings, fill assumptions, RC corner, coupling thresholds, net naming and hierarchical partitions.

A PRACTICAL DELAY ILLUSTRATION

NUMBERS ARE ILLUSTRATIVE

```
Driver + wire: Rwire = 180 ohm
Effective downstream C = 45 fF
Simple RC estimate R*C = 8.1 ps
(not exact Elmore/STA path delay)
```

Long nets and many vias also worsen RC; crosstalk can speed up or slow down transitions depending on aggressor direction. Do not equate a single lumped R*C product to the complete path delay.

EXTRACTOR / VERIFIER BOUNDARIES

StarRC	Synopsys signoff-grade parasitic extraction, generating parasitic views/files.
Quantus	Cadence extraction solution for interconnect RC and extraction integration.
Calibre	Siemens EDA verification families: DRC/LVS; extraction products depend on licensed stack.
IC Validator	Synopsys physical signoff verification with DRC/LVS capabilities.

Inputs/outputs: Extractor: routed geometry + process extraction tech → SPEF/RC database. Verifier: GDS/OASIS + netlist + rule decks → markers/error databases and compare results.

CROSS-PROBING A REAL NET

When one extracted net has unexpectedly high delay, locate its routed shapes and every via, compare reported R/C by segment and identify whether detour, wrong layer assignment or fill assumptions dominate.

THREE DIFFERENT VERIFICATION QUESTIONS

Design rule checking (DRC): are shapes legal under foundry geometry/manufacturing rules? Layout versus schematic (LVS): does extracted layout connectivity match intended netlist? Electrical rule checking (ERC): are specified electrical/connectivity rules satisfied? Passing any one does not imply passing the others.

TYPICAL FAILURE SIGNATURES

DRC spacing	Metal edges too close; inspect rule code, exact layer/corner, shapes, power straps and via enclosure.
LVS short/open	Two logical nets merged or one net disconnected; trace extracted netlist and layout labels.
LVS mismatch	Pin swaps, device/cell mapping, black-box setup, PG ties, wrong source netlist.
Missing SPEF	Corner/view mismatch, unmatched instance names, missing block parasitics or unannotated nets.

CORRELATION WORKFLOW

- Freeze routed database revision; align GDS/DEF, tech files, extraction corner and netlist names.
- Extract → load parasitics into PrimeTime/Tempus → verify annotation coverage → rerun max/min STA and SI.
- Keep tool deck revision and signoff waiver owners; document every remaining marker and severity.

WHY RESULTS DISAGREE

Implementation and signoff tools may differ in RC scaling, fill, corner mapping, coupling treatment or derates. Compare the same net/path, not just aggregate timing. Any correlation tolerance must be specified by the project methodology.

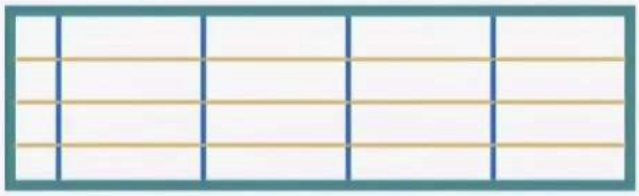
Exit gate: Signoff-quality extraction and clean or formally dispositioned DRC/LVS/ERC, with timing and power signoff consistent with the extracted layout.

A mismatching port may arise from incorrectly aliased supplies, hierarchy flattening, labels or inconsistent black-box definitions; inspect the first structural mismatch.

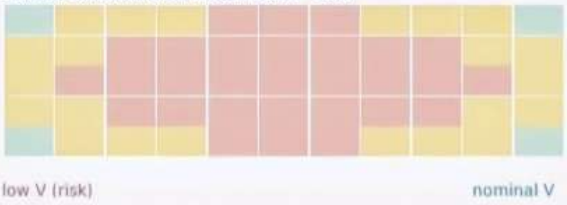
POWER INTEGRITY, EM/IR & FINAL SIGNOFF

Switching/leakage, voltage-drop analysis, power domains and physical reliability

PDN: RINGS, STRAPS AND CELL RAILS



ILLUSTRATIVE VOLTAGE DROP MAP



POWER GRID + VOLTAGE FIELD

Voltage map is illustrative, not measured silicon data

WHAT CONSUMES AND CARRIES POWER

Dynamic switching power is approximately P = activity × C × V² × f; short-circuit and internal cell components also contribute. Leakage persists without active switching and changes with process, voltage, temperature and cell selection.

Static IR drop studies quasi-steady loading; dynamic IR studies time-varying droop and local simultaneous switching. Supply drop weakens cell drive and can degrade timing; excessive ground bounce also matters.

ELECTROMIGRATION AND PDN

Electromigration (EM) is current-density/stress-driven metal wear-out. Assess wire/via current and technology-defined lifetime rules, not only absolute amperes. Thin straps, weak via arrays, hotspots and poorly connected macros raise risk.

TYPICAL REMEDIES / REPERCUSSIONS

- Reinforce straps, via arrays and PG cell hooks; reduce local current concentration and improve bump/pad feed planning.
- Spread highly switching logic where timing allows; investigate clock-tree power and unnecessary toggling.
- Recheck routing availability: wider PG consumes signal routing tracks and may amplify congestion.

UPF AND MULTIPLE POWER DOMAINS

Unified Power Format (UPF) describes power intent: supply sets, power domains, isolation, retention, power-state behavior and related level-shifter needs. Validate domain crossings and power-aware logical correctness; UPF is not a replacement for physical connectivity checks.

VOLTAGE DROP: AN IDEALIZED CALCULATION

SIMPLE STATIC ONE-PATH EXAMPLE

```
Local current I = 50 mA
Effective rail resistance R = 0.8 ohm
DC drop I*R = 0.040 V (40 mV)
At 0.90 V nominal -> ~0.86 V local
```

EM VISUALIZATION: CURRENT CROWDING

A narrow neck or single via concentrates current density. Wider legal metal and parallel via cuts reduce stress if connectivity and physical rules are satisfied.

TOOL ROLES AND INPUTS

Cadence Voltus	Power and rail/IR/EM analysis; consumes physical implementation, library/power/activity and rail data.
Ansys RedHawk-SC	Chip power-noise/reliability signoff analysis under compatible technology models.
Synopsys	PrimePower for power analysis; RedHawk-SC is Ansys. Synopsys rail-integrity options depend on licensed flow.

Vendor/tool selection varies by foundry-qualified methodology; all mentioned tools are not mandatory or interchangeable.

IR/EM DEBUGGING EXAMPLE

Symptom	Local low-voltage region near CPU/macro interface during a high-toggle pattern.
Check	Power-grid continuity, current source/bump assignment, activity provenance, density, via resistance, mode and analysis time.
Repair	Improve feeding straps/vias, repartition loads, validate macro PG connections; reanalyze both static and dynamic cases.

SIGNOFF MATRIX: DO NOT SKIP

- STA: setup, hold, recovery/removal, minimum pulse width, clocks, I/O, all active modes/corners.
- Geometry: route DRC, signoff DRC, LVS, ERC, antenna and required density/fill checks.
- Reliability: IR/EM, thermal assumptions, ESD/latch-up responsibilities if applicable to the scope.
- Integrity: final netlist/GDS/DB correlation, extracted RC, formal equivalence and complete run provenance.

Triage: A block is not "signed off" simply because WNS is positive. Closure requires all required signoff criteria and documented exceptions at the correct revision.

Always compare the predicted droop with the voltage range of the characterized timing views, rather than reusing nominal-voltage cell delay.

TCL, LINUX & ROOT-CAUSE DEBUGGING

Reusable scripts, log inspection, controlled design iterations and eight common failures



Reproducibility rule: fix one cause; retain baseline metrics, scripts, logs and database checkpoints.

STRUCTURED DEBUGGING LOOP

Save checkpoints and compare each hypothesis to the baseline

DAILY SHELL/EDA PROFICIENCY

LINUX; GNU GREP/FIND VARIATIONS EXIST

```
pwd; ls -lh; find . -name "*.rpt"
grep -nEi "error|warning|violated" run.log
tail -n 80 run.log; less timing.rpt
du -sh results/*; diff -u old.rpt new.rpt
git status; git diff -- scripts/
```

Use project naming conventions, separate logs/reports/results and never edit golden inputs in-place. Batch shells commonly accept a startup script (e.g., pt_shell -f run.tcl or icc2_shell -f flow.tcl; verify environment).

SMALL REPRODUCIBLE TCL EXAMPLE

PURE TCL; WRITES TWO TEXT FILES

```
set views {slow fast}
set prefix reports
file mkdir $prefix
foreach view $views {
    puts "Checking $view"
    set f [open "$prefix/$view.txt" w]
    puts $f "View name: $view"
    close $f
}
```

Line logic: set declares a list and output folder; file mkdir creates it; foreach iterates views; open/puts/close writes a simple report record. It does not configure STA corners automatically. Replace report writing with vendor API calls only after checking supported command syntax.

DESIGN OBJECT QUERIES

QUERY SEMANTICS ARE VENDOR-SPECIFIC

```
# Synopsys SDC-style collections
get_cells -hierarchical *
get_pins -of_objects [get_cells U1]
# Cadence Common UI: get_db uses its
  # own object model; not interchangeable
```

COMMON OPERATIONAL ERRORS

"Cannot link" often means library/hierarchy mismatch; "no valid clock" means SDC/clock object trouble; "unannotated parasitics" means SPEF/corner/naming mismatch. Debug inputs before adjusting optimization knobs.

Tcl branching: `if {$slack < 0} {puts "VIOLATION"}`. Reusable procedures use `proc name {args} {body}`; braces control Tcl substitution.

SYMPTOM → VERIFY → ACTION → OUTCOME

1. Congestion	Macro pin choke → examine overflow heatmap/blockage map → widen channel/move cells → reduced route overflow.
2. Setup WNS	Slow cell or wire → split path delay by arc/net and view → resize/localize/optimize → less negative max slack.
3. Hold WNS	Short min path/CTS imbalance → trace fast view and clock skew → legal delay insertion → improved min slack.
4. Clock skew	Unbalanced branch → sink latency/slew report → adjust tree routing/loads → controlled skew and clock slew.
5. Route DRC	Repeated local markers → analyze rule/layer cluster → pin access or route geometry fix → fewer markers.
6. RC gaps	Missing annotation → inspect SPEF map/names/corners → re-extract/repair mapping → higher coverage.
7. High IR	Sparse straps/hot load → inspect rail path and current map → strengthen PDN/spread load → better voltage margin.
8. Area growth	Buffer/size explosion → compare cell mix by stage → fix fanout/overconstraint → lower area without QoR loss.

METRICS TO LOG PER ECO

Configuration: git commit, netlist/SDC/PDK versions, tool build, corner/mode, run seed. QoR: WNS/TNS, violating endpoints, utilization, cell count/area, buffer count, worst slew/cap, DRCs, hotspot overflow, worst IR/EM. Capture delta and justification.

Golden debugging habit: Never claim a fix from a single improved report. Run full affected-view regression and document regressions and limits.

AUTOMATION ACCEPTANCE TEST

A rerun should recreate report filenames and quality metrics, fail early on missing input files, capture nonzero exit status and annotate deltas to baseline. Archive the most relevant path/DRC sample, not just the final score.

90-DAY ROADMAP, PORTFOLIO & INTERVIEWS

Practical open-source project, working-method discipline, 30/60/90-day job strategy



90-DAY LEARNING ROADMAP • measured deliverables at each gate, not only tool familiarity

FRESHER → CAPABLE JUNIOR ENGINEER

Study gates assume steady hands-on practice, not guaranteed mastery

BEFORE CHOOSING AN EDA COMMAND

Master Boolean logic, state machines, flip-flops, CMOS delay/power basics, Verilog/SystemVerilog, constraints (SDC), STA and the purpose of .lib/.lef. Learn design flow first; the commands become easier when you understand each stage's physical objective.

A REPRODUCIBLE 90-DAY SEQUENCE

Days 01–30	Write/simulate small RTL; learn setup/hold, SDC and reports; explain a failing max/min timing path.
Days 31–60	Synthesize and implement a small block; vary utilization, inspect congestion, clock tree and routing.
Days 61–90	Analyze final RC, compare two PPA runs, document design decisions and build interview-ready portfolio.

OPEN-SOURCE MINI-PROJECT PROPOSAL

Design: 8-bit counter or UART controller, then a modest pipelined datapath. Tools: Yosys + OpenROAD Flow Scripts (ORFS)/OpenSTA, KLayout; use redistributable Nangate45/SKY130 platform material. Inputs: RTL, clock/reset, SDC, platform libraries/LEFs.

ORFS DOCUMENTED EXAMPLE; FROM FLOW DIR

```
cd OpenROAD-flow-scripts/flow
make DESIGN_CONFIG=./designs/nangate45/gcd/config.mk
make gui_final
```

Deliverables: RTL test, synthesized netlist, placed/routed views, GDS, timing reports, utilization, cell area, congestion, documented DRCs, scripts and a before/after experiment. Run the official bundled GCD design first, then adapt a supported project configuration.

Measurable success: Reproducible clean run, no unexplained unconnected ports, documented timing/DRC status, and quantified impact of at least two controlled configuration changes.

Portfolio presentation: include one block diagram, the problem/constraint, toolchain, a failing report, a controlled change and quantitative before/after evidence. State any unresolved checks openly.

TECHNICAL INTERVIEW SELF-CHECK

- Why hold at fast PVT? Minimum-path delay shrinks; clock/skew relationships may make arrival too early.
- Why WNS worsens post-CTS? Clock propagation and updated skew change required time; placement and data paths may also change.
- Why not always upsize? It raises capacitance, area and sometimes routing load, threatening hold and power.
- DRC vs LVS? Geometric manufacturing legality versus intended connectivity equivalence.
- What is a valid ECO? Small controlled change with before/after timing, DRC, power, area and regression evidence.

FIRST 30 / 60 / 90 DAYS ON THE JOB

First 30	Learn build/scripts, PDK rules, release checks, owners, log conventions and bug escalation.
By day 60	Own small report/debug tasks; trace one full violation; propose evidence-backed ECOs.
By day 90	Own defined sub-block or closure category with review; track risks, revisions and signoff artifacts.

Survival under deadlines: communicate blockers early, preserve known-good databases, review corner/view coverage before overnight runs, automate comparison, avoid unreviewed constraints/waivers and never hide regressions.

TOOL SELECTION + CREDIBLE REFERENCES

Core skills: synthesis (DC/Genus/Yosys), place-route (ICC2/FC/Innovus/OpenROAD), timing (PrimeTime/Tempus/OpenSTA), RC (StarRC/Quantus), DRC/LVS (IC Validator/Calibre). Skills transfer, commands do not.

Public references: OpenROAD Flow documentation and tutorial (openroad-flow-scripts.readthedocs.io); Cadence Innovus datasheet (cadence.com); vendor documentation/release-specific help for proprietary commands.

INTERVIEW READINESS CHECKLIST

Explain one max and one min path numerically; identify SDC mistakes; interpret DEF/LEF/SPEF/GDS; distinguish formal, LVS and DRC; narrate a measured congestion or timing ECO.